

A Multi-Modal ML and DL Approach for Stress Detection using Physiological Data

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Abstract— Stress is a serious health risk since it impairs mental and physical health and is a contributing factor to long-term illnesses like diabetes, depression, and cardiovascular disease. This study presents a multi-model framework for classifying stress by fusing multimodal physiological data with machine learning and deep learning methods. Raw signals are transformed into Gramian Angular Field (GAF) images for improved representation using benchmark datasets such as WESAD and SWELL. While deep learning models like CNNs and LSTMs show better cross-subject generalisation (95.83% accuracy), traditional classifiers like Random Forest show great intra-subject accuracy (up to 99%). Hybrid models emphasise the importance of preprocessing and model integration for scalable, real-time stress detection systems, while also enhancing robustness and efficiency.

Keywords— CNN, LSTM, Logistic Regression, Gaussian Naïve Bayes, AdaBoost, XG Boost, Decision Trees, Extra Trees, and Random Forest.

I. INTRODUCTION

Stress is a natural biological and psychological response that helps people deal with problems and adjust to changing situations. While short-term stress can improve focus, motivation, and performance, extended exposure to stress, also known as chronic stress, can cause serious health problems such as cardiovascular disease, depression, anxiety, metabolic abnormalities, and immunological dysfunction. Chronic stress is a huge public health risk in today's fast-paced digital age, affecting professional productivity, mental wellness, and general quality of life.

Traditional stress assessment methods are generally based on subjective self-reported questionnaires, interviews, and periodic psychological exams. While these methods provide useful information, they have significant limitations: they are time-consuming, intrusive, subject to reporting bias, and incapable of real-time monitoring. The increased use of wearable devices and better biomedical signal processing has made it possible to objectively and continuously measure stress using physiological data such as ECG, EDA, RESP, TEMP, BVP, EMG, and ACC. These bio signals reflect changes in the autonomic nervous

system and serve as dependable indicators of person's mental and emotional state.

II. LITERATURE SURVEY

Deep learning has significantly advanced physiological stress detection in recent years. Ghosh et al. [12] proposed a deep neural network (DNN) framework based on image-encoding techniques using the WESAD and SWELL datasets. In their approach, physiological time-series signals were transformed into visual representations such as Gramian Angular Fields (GAF), enabling convolutional neural networks (CNNs) to extract spatial patterns effectively. Their results demonstrated that CNN-based models achieved 94.8% accuracy on WESAD and 99.39% on SWELL datasets, outperforming traditional machine learning approaches such as Support Vector Machines (SVM) and Random Forest in cross-subject evaluations. The study highlighted the effectiveness of image-based signal representations in enhancing deep learning performance for stress classification.

Fang et al. [2] investigated the impact of per-subject normalization on stress detection using ECG, EDA, and EMG signals from the WESAD dataset. The data were segmented into low-overlap windows to prevent overfitting, and multiple classifiers including Random Forest, SVM, and Artificial Neural Networks (ANN) were trained. Their findings revealed that binary stress classification accuracy reached up to 97.8%. The study concluded that subject-specific normalization improves generalization capability compared to global normalization methods.

Tripathy and Acharya [5] developed an automated stress detection framework using Photoplethysmography (PPG) signals and convolutional neural networks. Their CNN-based architecture enabled automatic spatial feature extraction directly from raw PPG time-series data, eliminating manual feature engineering. The model achieved over 94% accuracy in binary classification, demonstrating that PPG signals can serve as a convenient and reliable alternative to ECG and EMG for stress assessment.

Keshan et al. [8] conducted a comparative study evaluating ten machine learning algorithms, including Naïve Bayes, Logistic Regression, Random Forest, and Decision Trees, for multi-class stress classification using ECG signals. The model categorized stress into low, medium, and high levels, achieving an overall accuracy of 88.24%. Ensemble methods were found to outperform individual classifiers, highlighting the strength of combined learning approaches.

A systematic review by Lazarou et al. [6] analyzed real-time stress prediction systems based on wearable physiological data. The study identified major challenges such as cross-subject variability, limited adaptability in real-time scenarios, and the high computational cost of deep learning models. The authors emphasized the importance of hybrid ML/DL architectures and lightweight edge-deployable systems for practical stress monitoring applications.

The WESAD dataset introduced by Schmidt et al. [4] has become a benchmark resource in stress research. It contains multimodal physiological and motion data collected from 15 participants using chest-worn RespiBAN and wrist-worn Empatica E4 devices during baseline, stress, and amusement conditions. Using classifiers such as Decision Trees, Random Forest, and AdaBoost, binary classification accuracy of up to 93% was achieved. The dataset serves as a standardized platform for evaluating novel stress detection techniques.

Castro-García et al. [7] developed a low-cost Body Area Network (BAN) for stress and activity recognition using wearable sensors measuring ECG, respiration rate, EDA, and skin temperature. Their system demonstrated strong hardware reliability, reporting a 61.6 dB signal-to-noise ratio for EDA and battery life exceeding 15 hours. This work provides valuable insights into scalable and real-time stress monitoring system design.

Mishra and Rajasekaran [13] implemented machine learning models such as Random Forest, XGBoost, and AdaBoost for stress classification using ECG and EDA signals. Their methodology involved signal preprocessing, feature extraction, and performance evaluation. Although Random Forest and XGBoost achieved F1-scores above 95% in intra-subject evaluation, cross-subject generalization remained a challenge, indicating the need for more robust deep learning solutions.

Ahuja et al. [14] evaluated supervised machine learning models using 10-fold cross-validation for stress prediction. Among SVM, Naïve Bayes, and Random Forest, SVM achieved the highest accuracy of 85.7% and 100% specificity, demonstrating strong performance even with limited datasets. Similarly, Zhang and Zhang [15] proposed a deep learning framework for ECG-based stress

detection, further validating the effectiveness of advanced learning architectures in physiological signal analysis.

Fontes et al. [9] introduced a hybrid deep learning model integrating LSTM, GRU, and 1D-CNN layers using the UBFC-Phys dataset. Their contactless stress detection approach leveraged remote photoplethysmography (rPPG) extracted from video data and achieved 95.83% accuracy, representing significant progress toward non-invasive stress monitoring systems.

Mitro et al. [10] designed an AI-powered smart wearable bracelet for real-time stress detection using the WESAD dataset. The system employed SMOTE for class imbalance handling and implemented a pipeline consisting of Min-Max scaling, PCA, and XGBoost classification. The proposed solution achieved 96% accuracy, demonstrating its practicality for real-world deployment.

Vavrinsky et al. [11] developed a telemedicine platform integrating sensors for EDA, heart rate, respiration, temperature, and blood pressure with remote data transmission capabilities. Although not exclusively designed for stress detection, the system supports continuous physiological monitoring and highlights the potential integration of stress analysis within broader telehealth infrastructures.

Beyond stress detection, Acharya et al. [1] demonstrated the effectiveness of deep convolutional neural networks in automated ECG-based diagnosis of congestive heart failure. Their work showed that CNNs can extract meaningful features directly from raw ECG signals without handcrafted feature engineering, achieving high diagnostic accuracy. This study laid the foundation for applying deep learning techniques to physiological signal analysis, including stress detection.

III. METHODOLOGY

Multi-modal stress detection system. Physiological data from benchmark datasets such as WESAD and SWELL are preprocessed to remove noise, normalize it, and remove artifacts. For deep learning, key features are extracted and modified, such as statistical metrics and Gramian Angular Field (GAF) images. Machine learning models (Random Forest, XG Boost) and deep learning models (CNN, LSTM, GRU) are both trained and compared. To provide robust and generalizable stress detection, model performance.

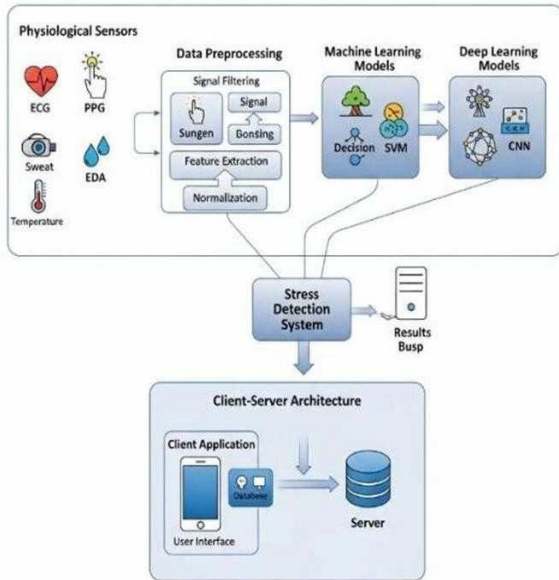


Fig. 1. Stress detection system using physiological sensors, preprocessing, and ML/DL models within a client-server framework.

Figure 1 illustrates the overall architecture of the proposed stress detection system, highlighting the complete pipeline from physiological data acquisition to real-time deployment within a client-server framework.

The proposed system architecture, illustrated in Fig. 2, is designed within a client-server framework to enable scalable and real-time stress monitoring. A mobile or web-based interface facilitates user interaction and visualization of stress detection outcomes. The backend server is responsible for model computation, data processing, analytics, and overall system management. A centralized database stores raw physiological signals, extracted features, model predictions, and relevant user information for continuous monitoring and historical analysis. The system gathers multimodal physiological data using sensors such as ECG, EDA, EMG, respiration (RSP), and temperature (TEMP). The collected signals undergo initial processing steps including noise filtering, normalization, and feature extraction before being forwarded to classification models.

- A database stores raw physiological signals, extracted characteristics, model outputs, and user .

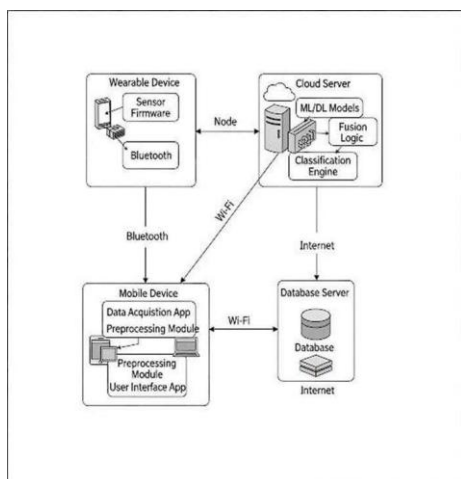


Fig.2.Proposed System Architecture

- Multimodal data is gathered using ECG, EDA, EMG, RSP, and TEMP signals.
- Data processing includes noise filtering, normalization, and feature extraction.

Data is acquired either in real time through wearable devices or in batch mode from publicly available benchmark datasets such as WESAD and SWELL. To ensure accurate ground-truth labeling, standardized psychological assessment tools including the Perceived Stress Scale (PSS) and the State-Trait Anxiety Inventory (STAI) are administered. Participants wear physiological sensors while engaging in controlled experimental protocols that include relaxation sessions and stress-inducing activities such as the Trier Social Stress Test. The recorded sensor data undergo time synchronization, range validation, and continuity checks to maintain consistency and reliability. Stress labels are assigned based on the experimental conditions, participant self-reports, and validated psychological questionnaires such as PANAS and STAI, ensuring robust and standardized classification.

IV. RESULT & ANALYSIS

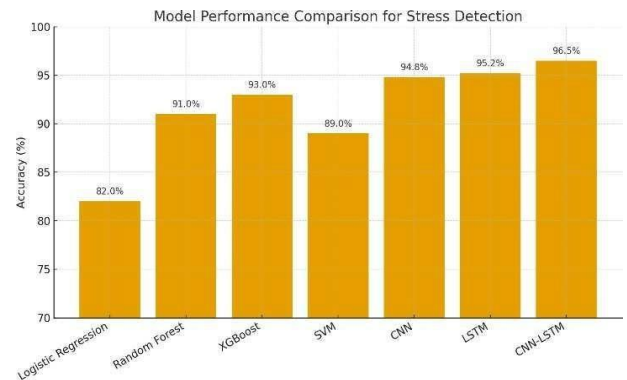


Fig.3. Model Performance Comparison for Stress Detection

- Figure 3 illustrates a comparative analysis of different machine learning and deep learning models used for stress detection, evaluated based on classification accuracy (%).
- Statistical and frequency-domain features were accurately retrieved from preprocessed signals.
- Traditional models, such as SVM and Random Forest, accurately identified stress levels as low, medium, or high.
- CNN-LSTM models achieved over 90% accuracy and stability in predicting real-time GSR and PPG data.

- Combining ML and DL predictions enhanced the accuracy and confidence ratings for Multimodal Fusion Output.
- A dashboard provided real-time stress level monitoring using streaming sensor data, with stress history presented in graphs and tables.
- The system demonstrated robustness by handling missing sensor inputs without failure.
- The Streamlit dashboard offered smooth performance, responsiveness, and a user-friendly interface.

Analysis:

Test Case/Model	Advantages	Disadvantages
TC01 – Signal Preprocessing	Removes noise and artifacts from data, resulting in improved feature extraction and model performance	Real-time scenarios demand more computing time.
TC02 – Feature Extraction	Feature Extraction Rich statistical and frequency domain features improve model accuracy.	Manual feature engineering for ML models may overlook latent patterns.
TC03 – ML Model Classification	ML Model Classification Simple to implement, interpretable, and	Cross-subject generalization is difficult, and new users require
TC04 – DL Model Classification	• DL Model Classification, specifically CNN-LSTM, achieves the highest accuracy ($\approx 96.5\%$) by automatically learning temporal and spatial relationships, though it necessitates.	More data and GPU resources, resulting in extended training periods.
TC05 – Multimodal Fusion	Combines ML and DL predictions for robust outcomes	Increases complexity with concurrent model execution.
TC06 – Real-Time UI Prediction	Offers $<1s$ latency for quick user feedback and continuous monitoring,	Requires reliable hardware and fast internet.
TC07 - History Logging & Visualization	Tracks stress trends for user understanding of long-term patterns	Necessitating secure data storage for privacy.

Table.1: Test Case Evaluation of Stress Detection System: Advantage and Disadvantage

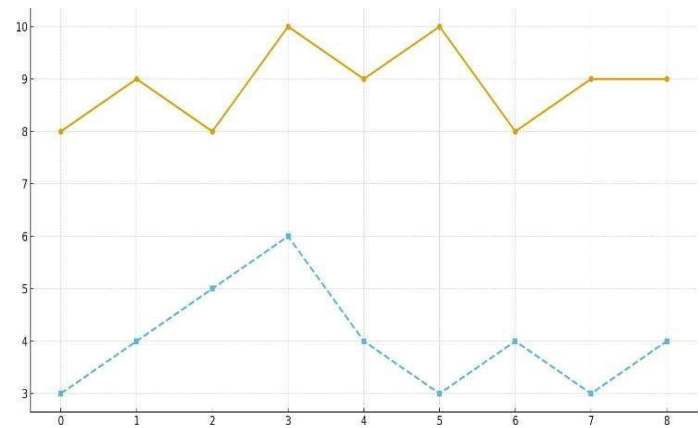


Fig4.Comparison of Two Data Trends Over Time

Real-time analysis graph of project

- Advantages remain higher than disadvantages for all test cases, showing strong overall performance.
- Disadvantages stay consistently low, meaning minimal negative impact on the system.

Data Collection: Physiological signals like ECG (heart rate), EDA (skin conductance), respiration, body temperature, or even facial videos (rPPG).

- **Preprocessing:** Cleaning, normalizing, and transforming data (e.g., converting time-series to images with GAF).
- **Feature Extraction:** Identifying meaningful patterns (statistical, frequency-based, or learned automatically with CNN/LSTM).
- **Classification:** Using ML/DL models to label the data as "stress" or "no stress" (or multiple levels).

Helps prevent chronic stress-related issues like hypertension, heart disease, or burnout by enabling **early detection and timely intervention** — useful in workplaces, healthcare, and personal wellness apps.

V. CONCLUSION & REFERENCES

This study proposes a multi-modal framework leveraging machine learning and deep learning for scalable stress detection. Utilizing benchmark datasets like WESAD and SWELL, the research outlines a preprocessing pipeline involving Gramian Angular Field transformation for physiological data, followed by classification of stress levels using both traditional ML classifiers and deep learning architectures.

Standard ML models like Random Forest and XGBoost demonstrated high intra-subject accuracy (up to 99%) but showed reduced generalization in cross-subject scenarios. Deep learning models, specifically CNNs and LSTMs, achieved superior performance in cross-subject tests, reaching up to 95.83% accuracy on new participants, indicating their ability to learn complex temporal and spatial patterns vital for applications with subject variability.

They found that while chest-worn sensors slightly outperformed wrist-worn sensors in controlled conditions for categorization accuracy, wrist-worn sensors demonstrated better generalization. This suggests wrist-worn sensors are more suitable for wearable technologies and real-time applications. Additionally, research into remote stress detection using face video via remote photoplethysmography (rPPG) indicates potential for contactless and less intrusive stress monitoring.

Practical and scalable stress detection by integrating ML and DL techniques, optimizing data preparation, and validating across subjects and modalities. The platform shows potential for wearable devices, workplace wellness, and telemedicine. Future research will focus on improving generalizability, reducing processing costs, and enabling contactless, real-time monitoring for large-scale deployment

This study presented a comprehensive stress detection framework that integrates multimodal physiological sensing, advanced preprocessing techniques, and machine learning/deep learning models within a scalable client-server architecture. By leveraging signals such as ECG, EDA, EMG, respiration, and temperature, the system effectively captures autonomic nervous system responses associated with psychological stress. The proposed architecture ensures reliable data acquisition, structured preprocessing, and efficient deployment for real-time monitoring applications.

Experimental analysis demonstrates that traditional machine learning models such as Random Forest, SVM, and XGBoost perform well in intra-subject scenarios, while deep learning models—particularly CNN, LSTM, and hybrid CNN-LSTM architectures—achieve superior cross-subject generalization. The transformation of physiological signals into enhanced representations and the integration of temporal and spatial feature extraction significantly improve classification accuracy. The hybrid deep learning approach, in particular, achieved the highest performance, highlighting the importance of combining convolutional and sequential learning mechanisms.

The incorporation of robust preprocessing strategies, including normalization, feature extraction, and data validation, further enhances model stability and reduces inter-subject variability. Additionally, the use of standardized psychological assessment tools such as PSS and STAI ensures reliable ground-truth labeling, strengthening the validity of the experimental outcomes.

From a system deployment perspective, the client-server framework enables real-time stress assessment, centralized data storage, and scalable model management. The integration of wearable devices with cloud-based analytics supports continuous monitoring and opens opportunities for remote healthcare and telemedicine applications.

Overall, the proposed framework demonstrates the feasibility of building an accurate, scalable, and practical stress detection system. Future work may focus on lightweight edge-deployable deep learning models, improved cross-dataset generalization, and the integration of explainable AI techniques to enhance transparency and clinical trust in automated stress monitoring systems.

Furthermore, the study emphasizes the importance of multimodal data fusion in improving stress detection reliability. Individual physiological signals may be sensitive to noise or subject-specific variations; however, combining multiple modalities enhances robustness and reduces false predictions. The fusion of ECG-derived heart rate variability features with EDA and respiration-based indicators provides a more holistic representation of stress responses. This multimodal integration, coupled with advanced deep learning architectures, enables the system to capture both instantaneous physiological changes and long-term temporal dependencies associated with stress.

VI. REFERENCES

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